

Name: Kunal Devidas Mahajan

Email: mahajankunal213@gmail.com

Title – Fake News Detector with LLM Explanation

1. Introduction

The rise of misinformation in the digital age has made fake news detection a critical task. In this project, we built an intelligent system that classifies news articles as real or fake, predicts the subject type (e.g., politics, entertainment), and provides a brief factual explanation using a large language model (LLM). The system combines traditional machine learning, deep learning, and transformer-based models, delivering end-to-end interpretability.

2. Abstract

This project presents a hybrid architecture that integrates Machine Learning (ML), Deep Learning (DL), and Natural Language Processing (NLP) to detect fake news. The workflow includes extensive text preprocessing, multi-class and binary classification models, and LLM-based explanation generation using FLAN-T5. A user-friendly interface was created using Gradio, allowing users to input any news article and receive a real-time prediction with explanation.

3. Tools and Technologies Used

Data Processing: pandas, numpy, re, spaCy

Machine Learning: scikit-learn, xgboost

Deep Learning: TensorFlow, Keras

Vectorization: TF-IDF, Tokenizer, pad_sequences

Language Model (LLM): Hugging Face Transformers – flan-t5-large

Interface & Deployment: Gradio, Google Colab, Hugging Face Spaces

Model Persistence: joblib, pickle

4. Steps Involved in Building the Project

a) Data Collection and Cleaning

Merged two datasets: Kaggle Fake News and GossipCop.

Cleaned titles, texts, dropped nulls, and standardized labels.

b) Preprocessing and Labeling

Combined title and text into a unified input.

Normalized text (lowercase, lemmatization, stopword removal).

Mapped 9+ noisy subject labels into 5 general categories: politics, world, general, national, entertainment.

c) Feature Extraction

For ML: Used TF-IDF on preprocessed text.

For DL: Used Keras Tokenizer + padded sequences (length 300).

Labels were encoded using Label Encoder.

d) Model Training and Evaluation

ML Models: Random Forest, Naive Bayes, and XGBoost (best performer: XGBoost with 94.4% accuracy).

DL Models: CNN, GRU, BiLSTM; CNN performed best with ~94.5% accuracy.

Real/Fake classification used a BiLSTM+CNN hybrid.

e) LLM-Based Explanation

Integrated flan-t5-large for generating brief, factual explanations.

Custom prompts were built using model predictions and news subject.

f) Deployment

Developed a Gradio app with three outputs:

Real/Fake Label

News Subject

Explanation (via LLM)

Hosted via Google Colab with plans to migrate to Hugging Face Spaces.

5. Conclusion

This project demonstrates how classical ML, DL, and LLMs can work in harmony to not only detect fake news but also provide interpretability. The Gradio interface allows end users to interact with the system intuitively. In future iterations, real-time fact-checking APIs or browser extensions can be added to make this a robust misinformation tool.

Live Demo App: [APP LINK](#)

Demo video: [Video Link](#)